THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** two interacting objects	Forces are interactions
2	**(d)** balanced	Net force is zero
3	**(b)** irregularities of surfaces in contact	Even smooth surfaces have tiny bumps
4	**(c)** repel	Magnetic force
5	**(b)** positive and negative	Like repel, unlike attract
6	**(c)** 60 N/m²	Pressure = force ÷ area
7	**(d)** the area is small so pressure is large	Same force on small area
8	**(b)** increase water pressure at taps	Greater height gives greater pressure
9	**(c)** pressure inside our body balances it	Internal pressure equal to outside
10	**(d)** lower air pressure	Bernoulli-type effect

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Earth's gravitational force pulls it down (1). Gravity is always attractive (1).	1+1
12	Mass is amount of matter in kg and is constant (1). Weight is force of Earth's pull in newtons and varies (1).	1+1
13	Air moves from high pressure to low pressure region (1). Flow continues until pressures equalise (1).	1+1
14	Wind from sea to land by day (1). Land heats faster so low pressure forms over land (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Push an empty bottle into water (1). Feel an upward push (1). It bounces back to the surface due to upthrust (1).	1+1+1
16	Warm ocean water heats air which rises (1). Vapour condenses and releases heat, lowering pressure (1). Earth's rotation makes the winds spin around the low pressure (1).	1+1+1
17	If gravitational force is greater, object sinks (1½). If equal, object floats (1½).	1½+1½
	OR: Formed from lava with trapped gas bubbles (1). Has air pockets so is porous (1). Less dense than water so floats (1).	1+1+1
18	Eye at centre and strong winds around it (1½). Eye has the lowest pressure and calm wind (1½).	1½+1½
	OR: Storm surge floods coasts (1). Heavy rain causes floods and landslides (1). Salt water ruins farmland and drinking water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Earth: 20 N (1). Moon: 3.2 N (1). Mars: 7.6 N (1). Weight depends on planet's gravity (1). Mass remains 2 kg (1).	1+1+1+1+1
	OR: Thrown upwards, moves straight (1). Speed decreases while going up (1). Momentarily stops at the top (1). Speed increases coming down (1). Because gravity always pulls downward (1).	1+1+1+1+1
20	Warm ocean water heats air (1). Rising moist air condenses (1). Heat is released lowering pressure (1). Earth's rotation causes spin (1). Eye at centre, strong winds around (1).	1+1+1+1+1
	OR: Storm surge and floods (1). Heavy rain, landslides (1). Salt ruins soil (1). Early warning and evacuation (1). Strong shelters (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** kilograms	Amount of matter
2	**(d)** 1.6 N	Weaker gravity
3	**(a)** 0.2 N	1 ÷ 5
4	**(d)** greater than buoyant force	Net force downward
5	**(b)** chewing food and pumping blood	Inside and outside the body
6	**(c)** heats faster than water	Creates low pressure
7	**(d)** sudden flow of built-up static charges	Air insulating property breaks down
8	**(d)** crouch low in an open area away from tall objects	Minimise contact with ground
9	**(c)** moist air supply is cut off	Needs warm moist air
10	**(d)** Kerala, Karnataka and Tamil Nadu	Help mangoes ripen

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	3 × 10 (1). 30 N (1).	1+1
12	Measures weight or force (1). Range is the maximum weight it can measure (1).	1+1
13	Fast air lowers pressure between them (1). Higher outside pressure pushes them together (1).	1+1
14	Fast wind makes low pressure above the roof (1). Higher pressure inside pushes the roof up (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mass is amount of matter and remains same (1½). Weight depends on gravitational pull of the planet (1½).	1½+1½
16	Warm moist air supply from ocean is cut off (1½). So energy source declines and strength reduces (1½).	1½+1½
17	Lifting a bag (1). Chewing food (1). Heart pumping blood (1).	1+1+1
	OR: Walking (1). Holding objects (1). Brakes (1).	1+1+1
18	Kalboishakhi – West Bengal (1). Bordoila – Assam (1). Mango showers – Kerala, Karnataka (1).	1+1+1
	OR: To increase the area (1). To decrease pressure on shoulders (1). To carry heavier loads comfortably (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Magnetic: attract or repel; compass (1½). Electrostatic: attract or repel; rubbed balloons (1½). Gravitational: attractive only; falling apple (2).	1½+1½+2
	OR: Upthrust equals weight of liquid displaced (1). Ship displaces a large volume (1). So upthrust equals ship's weight (1). Nail displaces little (1). Upthrust is smaller than its weight so it sinks (1).	1+1+1+1+1
20	Less height gives lower pressure (2). P = F/A (1). 900 ÷ 0.3 (1). 3000 N/m² (1).	2+1+1+1
	OR: Thunderstorm over land (1). Cyclone over warm ocean (1). Cyclone has eye (1). Cyclone has stronger winds and storm surge (1). Thunderstorm has lightning and thunder (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET C

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** change the mass of an object	Mass is fixed
2	**(d)** friction	Needs touching surfaces
3	**(c)** rough surface	More irregularities
4	**(d)** attract	Electrostatic force
5	**(c)** attractive	Never repels
6	**(b)** 300 N/m²	Pressure = force ÷ area
7	**(d)** smaller area gives more pressure	P = F/A
8	**(c)** pressure is largest near the bottom	Horizontal pressure increases with depth
9	**(a)** outside air pressure is more than inside	Air pushed out
10	**(d)** greater	Larger difference drives faster air

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Force exerted by a liquid on an object (1). Upward direction (1).	1+1
12	A body fully or partly immersed in a liquid experiences an upward force (1). It equals the weight of liquid displaced (1).	1+1
13	Charges separate in clouds by rubbing of ice and water (1). They discharge suddenly when air insulation breaks (1).	1+1
14	Lightning heats air rapidly (1). Air expands suddenly and makes a loud sound (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ship displaces much water (1). Upthrust equals its weight (1). Nail displaces little water so sinks (1).	1+1+1
16	Millibar or hectopascal (1). 1 hPa = 100 Pa (1). 101300 Pa (1).	1+1+1
17	Upthrust = weight in air − apparent weight (1). 8 − 6 (1). 2 N (1).	1+1+1
	OR: No, it acts without contact (1½). Earth pulls objects falling towards it even from a distance (1½).	1½+1½
18	Fast wind over roof lowers pressure (1). Air inside is at higher pressure (1). Net upward force can lift the roof (1).	1+1+1
	OR: Salt water enters farmland (1½). Salt makes the soil less fertile (1½).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hang stone in air and note weight (1). Lower it into water (1). Note reduced reading (1). Difference gives upthrust (1). Upthrust acts upward (1).	1+1+1+1+1
	OR: Hand and table interact (1). Magnet and iron interact (1). Earth and apple interact (1). Force needs two objects (1). Each exerts force on the other (1).	1+1+1+1+1
20	Kalboishakhi in West Bengal, Bihar, Jharkhand (1½). Bordoila in Assam (1). Mango showers in southern states (1). Help kharif crops, mangoes and coffee (1½).	1½+1+1+1½
	OR: 0.1 m²: 400 N/m² (1½). 0.05 m²: 800 N/m² (1½). Smaller area means more pressure (2).	1½+1½+2

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ANSWER KEY & MARKING SCHEME SET D

Chapters 5 & 6: **Exploring Forces, Pressure, Winds, Storms and Cyclones** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** newtons	A force
2	**(a)** Jupiter	25.4 N
3	**(d)** 0.2 N	2 ÷ 10
4	**(d)** weight of liquid displaced	For fully or partially immersed object
5	**(c)** liquids and gases	Fluids oppose motion
6	**(c)** move towards each other	Low pressure between
7	**(b)** lightning heats and expands air	Rapid expansion
8	**(c)** metal rod that carries charge to the ground	Protects buildings
9	**(c)** storm surge	3–12 m high
10	**(d)** Assam	Pre-monsoon

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Floats when weight equals upthrust (1). Sinks when weight is greater than upthrust (1).	1+1
12	Water exerts an upward force (1). It reduces the effort needed to lift it (1).	1+1
13	Region of lowest pressure at the centre (1). Calm winds there (1).	1+1
14	Stay away from tall objects and crouch low (1). Avoid metal umbrellas and get out of water (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Water fills irregularities (1). Friction reduces (1). Less grip for feet (1).	1+1+1
16	P = F/A (1). 60 ÷ 0.2 (1). 300 N/m² (1).	1+1+1
17	Mass in kg, weight in N (1). Mass constant, weight varies (1). Mass is amount of matter, weight is a force (1).	1+1+1
	OR: Diagram (1). Spring and hook (1). Scale in newton and gram (1).	1+1+1
18	It increases area of contact (1½). It reduces pressure on the head (1½).	1½+1½
	OR: Between balloons pressure falls (1). Over house roofs pressure falls (1). Hence roofs may be lifted (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pull a block with spring balance on a rough surface (1). Note reading (1). Repeat on a smooth surface (1). Rough surface needs more force (1). Greater friction on rough surface (1).	1+1+1+1+1
	OR: Gravity pulls the swing down (1). Muscular force from push starts the motion (1). Friction at the pivot and air slows it (1). At highest point speed is zero momentarily (1). Direction of motion changes continuously (1).	1+1+1+1+1
20	Press out air (1). Pressure inside decreases (1). Outside pressure is higher (1). Force needed to overcome difference (1). Releasing allows air in (1).	1+1+1+1+1
	OR: Pressure differences cause wind (1). Sea and land breezes (1). Storms from low pressure over heated land (1). Cyclones from very low pressure over ocean (1). Speed larger with greater difference (1).	1+1+1+1+1

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